CERTMEM: EVIDENCE RESIDUALS FOR TOKEN-EFFICIENT AGENT MEMORY

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Manuscript in preparation

ABSTRACT

Long-term conversational agents must retain detailed histories while limiting the tokens spent maintaining and consulting memory. Existing systems connect compact summaries to original evidence, enabling selective recovery of information lost during compression. However, recoverability alone leaves a coupled allocation problem: which details should be preserved in compact memory, and which should be recovered when a question arrives? We propose CertMem, a framework that uses a shared record of compression omissions to coordinate these decisions. At write time, CertMem audits candidate summaries against source-grounded checks and retains a small set of verbatim evidence residuals, prioritizing their predicted reduction in downstream answering loss per added token. At read time, the same audit guides selective access to additional source spans, with reassessment after each expansion. The controller is designed to learn from paired answering outcomes, while independent validation separates useful diagnostic scores from claims of answer correctness. We formulate memory efficiency in terms of total maintenance and query-time token expenditure and specify matched-budget evaluation on LoCoMo and LongMemEval. The central hypothesis is that preserving the right omissions before retrieval can reduce repeated source reading and improve the accuracy–cost frontier beyond summary-based routing alone.

1 INTRODUCTION

A conversational agent may remember that a user discussed a conference in Kyoto yet forget that the trip was canceled. A later question about the user's travel plans then requires a distinction that the stored summary no longer expresses. Retaining the original conversation makes the cancellation recoverable, but consulting the entire discussion for every related question spends tokens on information that the summary already preserves. Long-term memory therefore requires decisions about both what to retain in compact form and how much original evidence to read. Benchmarks such as LoCoMo and LongMemEval make these decisions consequential by testing facts, temporal relations, and updates across extended interaction histories (Maharana et al., 2024; Wu et al., 2025).

Recent work provides increasingly effective connections between abstraction and evidence. RMM stores topic summaries paired with raw dialogue and adapts memory ranking using citation feedback (Tan et al., 2025). HiGMem organizes events above dialogue turns, using event summaries and fact sheets to guide LLM selection of the turns that support an answer (Cao et al., 2026). H-MEM routes through domain, category, memory-trace, and episode layers using indices to subordinate memories (Sun et al., 2026). SEEM combines relational facts with episodic event frames and uses reverse provenance expansion to recover associated source passages (Lu et al., 2026). Most directly, TierMem answers from summaries when sufficient, escalates to linked raw pages otherwise, and can consolidate verified findings for subsequent queries (Zhu et al., 2026). These advances establish hierarchical retrieval, provenance links, and selective source access as foundations on which our work builds.

They also motivate a more specific question: how should information about *what compression removed* influence both retention and retrieval? A source pointer identifies where a memory came from; it does not, by itself, describe which distinctions survived its summary. In the Kyoto example, a brief cancellation utterance may be more valuable than a longer, fluent description of the conference. A system can retain that utterance alongside the summary or recover it during a future

query. The two choices have different costs that depend on how often the information is needed. We hypothesize that explicitly recording omissions can improve this allocation by letting the writer preserve inexpensive, consequential details and informing the reader about unresolved gaps. The hypothesis concerns how compression diagnostics guide both stages.

Evaluating this hypothesis requires distinguishing summary fidelity from task performance. Preserving many extracted facts can still leave an answer wrong when a missing condition, attribution, or temporal relation determines the result. Conversely, a detail omitted from a summary need not matter to the questions the agent receives. TRUSTMEM already evaluates memory transitions through coverage, preservation, and faithfulness (Yang et al., 2026), while DeMem formulates memory quality through decision loss induced by compression (Zou et al., 2026). Our proposed mechanism connects such diagnostic information to a concrete intervention: selecting short source passages that repair predicted answering failures and reusing the remaining gaps to guide later evidence access. Evaluation must include downstream failures that the diagnostic checks did not anticipate.

We propose **CertMem**, a framework for coordinating memory compression, evidence retention, and source expansion under token budgets. For each source segment, a writer produces candidate summaries at several lengths. A bounded audit checks source-grounded propositions and selected dependencies, including conditions, negations, and temporal qualifiers. Its output records which checked items are retained, contradicted, or unresolved and links them to supporting source spans. CertMem then selects a compact *evidence residual*: verbatim passages intended to repair the most consequential omissions. Selection uses predicted reduction in answering loss per added token and reassesses overlapping candidates as the residual grows. When a reference or condition requires several passages jointly, the candidate is a passage bundle. The resulting memory contains the summary, selected residual, source links, and a bounded audit record; the audit describes inspected content rather than certifying exhaustive coverage.

At query time, a retriever first supplies compact memories and their residuals. The controller relates the query to the recorded gaps and selects additional source spans when their predicted benefit justifies the added cost. Each expansion is followed by reassessment, so reading part of a source does not automatically mark all its omissions as resolved. A bounded search over the raw archive remains available when relevant evidence lies outside the retrieved summary links. The controller stops when evidence passes a validation-selected threshold or the read budget is exhausted, recording unresolved cases. Development uses paired answers from compact and expanded contexts with a fixed reader. Future benchmark questions are excluded from memory construction, and policy selection uses conversations separate from the final evaluation. Neither an entailment score nor a stopping decision is treated as a guarantee that an individual answer is correct.

Our optimization target makes the full cost of this procedure explicit. For a history H, queries $q_{1:N}$, and a write-and-read policy π, we consider

$$\min_{\pi} \mathbb{E}\left[C^{\pi}_{\text{maint}}(H) + \sum_{j=1}^{N} C^{\pi}_{\text{query}}(H, q_j)\right] \quad \text{subject to} \quad R_{+}(\pi) \leq \epsilon. \tag{1}$$

Here, maintenance includes compression, auditing, and residual construction; query cost includes routing, evidence inspection, and answer generation. R_{+} is the expected positive increase in task loss over a fixed source-based reference, using the same ground truth. Absolute accuracy is also required because the reference can fail. Equation 1 specifies a target rather than an established guarantee. Persistent storage, including the raw archive and metadata, is reported separately.

We outline evaluation on LoCoMo and LongMemEval under matched reader models, data splits, and token budgets. The decisive comparisons are with summary-only memory, raw retrieval, the five preceding frameworks, and controlled variants that remove residuals or replace their selection with relevance or fact-coverage scores. Accuracy–cost curves, false stopping decisions, and repeated-query workloads test whether proactive retention offsets its maintenance overhead. The proposed contributions are: (i) a formulation linking write-time retention and query-time evidence allocation through recorded compression omissions; (ii) an evidence-residual mechanism that uses these omissions for both selective preservation and source expansion; and (iii) an evaluation protocol that tests this mechanism through downstream answer quality and complete token accounting. Empirical improvements remain to be measured.

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